

Multiplexing & De-multiplexing

Efficient transmission of multiple signals over a single communication channel

Topic

Unit 3 — Data Link & Transport Layer

Subject

Computer Networks

Technique

MUX / DEMUX Architecture

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Introduction to Multiplexing

Definition

Multiplexing is a technique in which multiple analog or digital signals are combined and transmitted simultaneously over a single shared communication channel. This maximizes the use of expensive or limited transmission infrastructure.

Why Do We Need Multiplexing?



Cost Efficiency

Laying separate physical cables for every communication pair is expensive. One shared link reduces infrastructure costs dramatically.



Bandwidth Utilization

A single medium like fiber optic or coaxial cable has enormous bandwidth that would be wasted if only one signal used it at a time.



Scalability

Networks can support more users without adding new physical links — simply add more multiplexed channels on the existing medium.

MUX System Architecture & Working

Step-by-Step Working of a Multiplexed System:

1

Signal Sources

Multiple senders (n signals) — voice, video, data — are fed as inputs to the MUX device.

2

Multiplexer (MUX)

The MUX combines all n input signals into one composite signal using FDM, TDM, or WDM technique.

3

Transmission Medium

The single composite signal travels across the shared link — coaxial cable, twisted pair, or fiber.

4

De-multiplexer (DEMUX)

The DEMUX at the receiving end separates the composite signal back into n individual signals.

5

Destination Receivers

Each separated signal is delivered to its correct destination application or device.



Key Terminology

Channel

The physical medium shared among multiple senders.

Bandwidth

Total frequency capacity of the channel (Hz or bps).

MUX

Multiplexer — combines signals at sender side.

DEMUX

De-multiplexer — separates signals at receiver.

Composite Signal

The single combined signal traveling on the shared link.

Guard Band

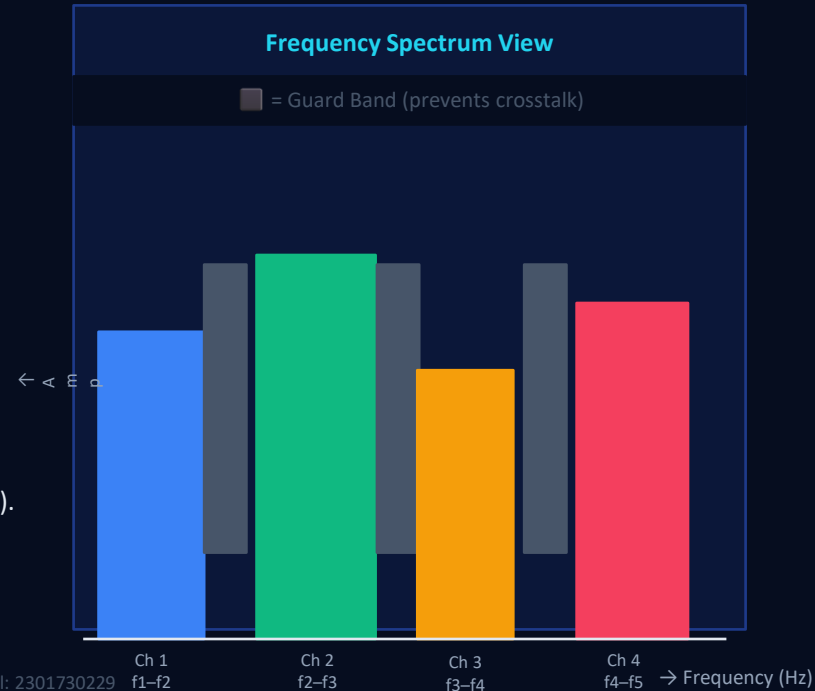
Unused frequency gap between channels to prevent crosstalk in FDM.

Frequency Division Multiplexing (FDM)

In FDM, the total available bandwidth of the channel is divided into smaller, non-overlapping sub-bands (sub-channels). Each sender is assigned a unique frequency band and transmits continuously and simultaneously with all other senders.

Detailed Working:

1. Each signal source is assigned a dedicated carrier frequency ($f_1, f_2, f_3, \dots$).
2. Each signal modulates its carrier to shift it to the assigned frequency band.
3. All modulated sub-channels are combined (summed) into one composite signal by the MUX.
4. The composite signal is sent over the shared transmission medium.
5. At the receiver, band-pass filters separate each sub-channel by its frequency range.
6. Each sub-channel is demodulated to recover the original signal.
7. Guard bands (unused frequency gaps) prevent inter-channel interference (crosstalk).



Time Division Multiplexing (TDM)

In TDM, the channel is divided into time slots rather than frequency bands. Each user is given the entire bandwidth of the channel, but only for a very short, repeating time slot within a frame. Signals take turns using the medium.

Frame Structure — Time Slot Allocation:



Synchronous TDM vs Statistical (Async) TDM:

Synchronous TDM

- Fixed, pre-assigned time slots per device
- Slot assigned even if device has NO data
- Leads to bandwidth wastage (empty slots)
- Simple and easy to implement
- Used in: T1/E1 telephone carrier lines, SONET

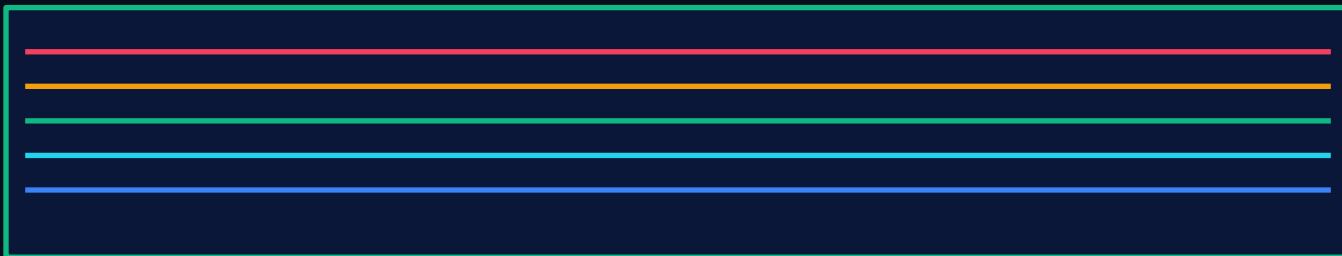
Statistical (Async) TDM

- Slots dynamically assigned only to active devices
- No slot wasted — bandwidth used efficiently
- Requires address/ID field in each slot
- More complex but much higher efficiency
- Used in: ATM networks, modern broadband

Wavelength Division Multiplexing (WDM)

WDM is the optical fiber equivalent of FDM. Instead of frequency bands, different wavelengths (colors) of light are used to carry different signals simultaneously through a single optical fiber. WDM dramatically increases the capacity of fiber optic networks.

WDM Optical Fiber — Light Wavelength Channels:



Single Optical Fiber (all λ travel together)

λ_1 (1310nm)

λ_2 (1430nm)

λ_3 (1550nm)

λ_4 (1570nm)

λ_5 (1590nm)

WDM vs DWDM:

WDM uses wide channel spacing (~20nm apart). Supports ~8–16 channels. Used in shorter distances and metro networks.

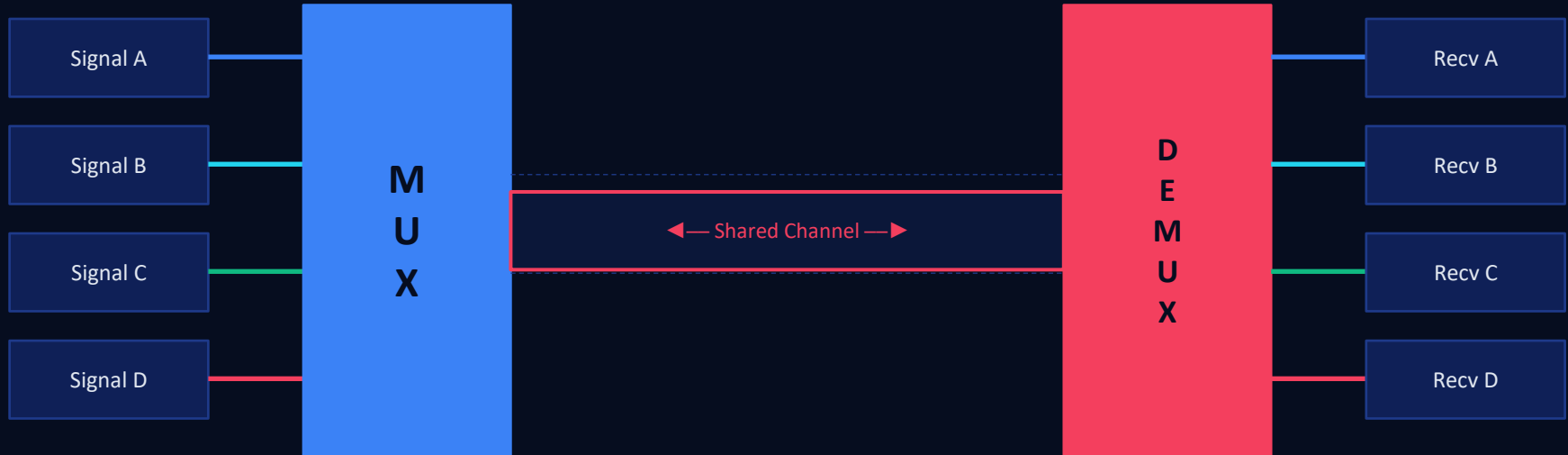
Dense WDM (DWDM):

DWDM uses very narrow spacing (~0.8nm apart). Supports 40–160+ channels on one fiber. Used in long-haul backbone networks (transoceanic cables).

De-multiplexing — Concept, Working & Types

De-multiplexing is the reverse operation of multiplexing. The De-multiplexer (DEMUX) receives the single composite signal from the shared channel, identifies each individual signal using frequency, time slot, or code information, and routes each to its correct destination receiver.

Complete MUX → Transmission → DEMUX Architecture:



In FDM:

Band-pass filters at the DEMUX select specific frequency ranges to separate each channel.

In TDM:

DEMUX reads the time slot number/sequence to route each slot to the correct output.

In WDM:

Optical filters (prisms/gratings) separate different wavelengths to different output fibers.

Multiplexing in the Transport Layer

At the Transport Layer (Layer 4), multiplexing means: a single host can run multiple applications simultaneously and each application's data is multiplexed into the network. Port numbers act as the multiplexing identifiers — they tell the Transport Layer which application a segment belongs to.

Port Number Concept — Socket = IP Address + Port Number:

0 – 1023	Well-Known Ports HTTP(80), HTTPS(443), FTP(21), SSH(22), DNS(53), SMTP(25)
1024–49151	Registered Ports MySQL(3306), PostgreSQL(5432), RDP(3389), MongoDB(27017)
49152–65535	Dynamic / Ephemeral Assigned temporarily by OS for client-side connections

TCP vs UDP — Transport Layer Protocols:

TCP — Transmission Control Protocol

- Connection-oriented (3-way handshake: SYN → SYN-ACK → ACK)
- Reliable, ordered, error-checked delivery guaranteed
- Flow control (sliding window) & congestion control
- Higher overhead — slower but dependable
- Applications: HTTP/S, FTP, SSH, Email (SMTP/IMAP)

UDP — User Datagram Protocol

- Connectionless — no handshake or session setup
- No guarantee of delivery, order, or error recovery
- Very low overhead — extremely fast transmission
- Application must handle reliability if needed
- Applications: DNS, VoIP, Live Streaming, Online Gaming

Comprehensive Comparison of MUX Techniques

Feature	FDM	TDM	WDM	CDM
Domain Used	Frequency	Time	Wavelength (Light)	Code (CDMA)
Signal Type	Analog	Digital / Analog	Optical	Digital
Sharing Basis	Frequency Band	Time Slot	Light Wavelength	Unique Code
Guard Bands	Required	Not Needed	Not Needed	Not Needed
Bandwidth Use	Fixed per channel	Whole BW per slot	Very high (fiber)	Full BW, coded
Efficiency	Moderate	High	Extremely High	High

 **Real-World Applications:**

Cable TV → FDM PSTN / GSM → TDM Fiber Backbone → WDM/DWDM 4G LTE → OFDM (advanced FDM) 5G NR → OFDM + TDM hybrid

Key Takeaways

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01

Multiplexing

Allows efficient sharing of a single channel among multiple senders — saving cost and maximizing bandwidth utilization.

02

FDM

Divides bandwidth into frequency bands. Used in analog systems like FM radio and cable TV. Needs guard bands.

03

TDM

Divides transmission time into frames/slots. Two types: Synchronous (fixed slots) and Statistical (dynamic slots).

04

WDM / DWDM

Optical fiber technique using different wavelengths of light. DWDM supports 40–160+ channels per fiber.

05

De-multiplexing

Reverse of MUX — separates composite signal at receiver. Uses filters (FDM), time counters (TDM), or optical prisms (WDM).

06

Transport Layer

TCP & UDP use port numbers for multiplexing. Socket = IP + Port. Well-known ports: HTTP(80), DNS(53), FTP(21).

Thank You!

Questions Welcome 🙌